QP CODE: 26900531

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION OCTOBER 2026
FIFTH SEMESTER
DISCIPLINE SPECIFIC ELECTIVE (DSE) - MG5DSESTA302 – STATISTICAL COMPUTING USING PYTHON

(2024 ADMISSION ONWARDS)

SCHEME FOR VALUATION

Duration: 1 hour 30 minutes **Maximum Marks: 50**

PART A
Short answer type questions
Answer any 7 questions. Each question carries 2 marks.

1. Syntax of if-elif-else statement and example - 1mark each
2. Any two advantages of python IDE / (definition) - 2 mark
3. Definition of logical error and syntax error - 1 mark each
4. Python code for removing missing values from a Data Frame - 2 mark
5. Python code for inner join of two datasets - 2 mark
6. Python code to create an array of five zeros - 2 mark
7. Definition of relplot and scatterplot - 1 mark each
8. Definition of swarmplot and boxplot - 1mark each
9. Interpretation of QQ plot / definition of QQ plot - 2 mark
10. Explanation of inverse transform method - 2 mark

PART B
Short essay type questions
Answer any 4 questions. Each question carries 6 marks.

11. Explanation of any three python datatypes with example - 2 mark each
12. Describe any 4 built-in functions in python with example - 6 mark
13. Creating a long format sales data frame - 1 mark
    Python code for obtaining total sales using pivot table and output - 2 mark
    Python code for melt the data frame back to long format - 1 mark
    Drawing of matplotlib bar chart of total sales and output - 2 mark
14. Code for creating data frame from a list of tuples - 2 mark
    Code for data frame shape, columns and data types with output - 4 mark

15. Importing libraries and load the titanic dataset - 2 mark
Code for creating boxenplot - 3 mark
Output and interpretation of results - 1 mark
16. Detailed explanation of ONE WAY ANOVA - 6 mark

PART C
Essay type questions
Answer any 1 question. Each question carries 12 marks.

17. Importing necessary libraries - 1 mark
Defining appropriate data structures for categories and corresponding values - 2 mark
Python Code for Pie Chart representation - 3 mark
Python Code for Bar Chart representation - 3 mark
Adding appropriate titles, axes labels and adjustments - 2 mark
Correct use of plt. subplots and plt. show to render the visualization on screen - 1 mark
18. (a) Definition of sphericity - 2 mark
(b) Python code for Mauchly's test of sphericity and output - 5 mark
(c) python code for repeated measures ANOVA - 3 mark
ANOVA Results and interpretation - 2 mark

Final mark summary

Part	Valuation	Maximum
A	Any 7 × 2	14
B	Any 4 × 6	24
C	Any 1 × 12	12
Total		50